

MOHAK DUA

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EDUCATION

University of Minnesota Twin Cities

Bachelor of Science in Computer Engineering (GPA: 3.92)

Expected May 2028

EXPERIENCE

Calyan Technologies Inc.

Electrical Engineering Intern

May 2026 – August 2026

- Designed a 2-layer USB-to-UART programmer board in Altium to flash an internal ESP32 system, routing the USB differential pair and a dual-transistor auto-boot circuit around a CP2102N bridge. Assembled and bench-debugged it, cutting flashing to one step. Scaled to 20+ units as the team's standard tool.
- Modeled a high-voltage capacitor-charging chain in LTspice across its boost, flyback, and HV bank stages. Traced a charge failure to a 5x power gap between the boost supply and the flyback load. Flagged the defect before fabrication.
- Completed 120+ requirements-based firmware and electrical verification and validation (V&V) steps. Revised the test protocol, documented defects and results, and automated current measurements with Python.
- Built an ESP32 system that collected synchronized data from three accelerometers at a stable 400 Hz. Developed the firmware and Python tools for live plotting, filtering, and logging.

UMN Solar Vehicle Project and Gopher Motorsports Formula SAE

Hardware Design Engineer

September 2024 – Present

- Designed a 2-layer power-distribution board in Altium that gated 12V vehicle power to the Raspberry Pi UI through a high-side P-channel MOSFET load switch and regulated a 5V rail with a TPS54560 buck converter, with input fusing and transient protection.
- Built a 4-layer STM32 test board in Altium that reproduced the car's sensor interfaces with buffered analog channels, DAC-driven outputs, PWM and frequency generators, and a CAN transceiver for hardware-in-the-loop validation of the Sensor Hub.
- Assembled wiring harnesses, soldered and inspected boards, and root-caused electrical faults across both teams, including live debugging at competition drive days.
- Programmed STM32 firmware that drove calibrated analog, PWM, and CAN signals, injected out-of-range and dropout faults, and replayed logged drive-day data to reproduce field bugs on the bench across firmware and hardware revisions.

Meerkats World

Embedded Systems Intern

May 2025 – August 2025

- Prototyped a waterproof IoT monitoring system for heat-sink manufacturing, integrating an ESP32 with pH, TDS, and temperature sensors across analog and digital interfaces.
- Calibrated sensors against control solutions and built a statistical validation model in Excel to characterize sensor drift and hysteresis.
- Deployed an MQTT telemetry pipeline to a real-time Grafana dashboard, and validated reliability through 72-hour continuous stress testing of enclosure integrity and firmware stability.

PROJECTS

FPGA Mixed-Signal Oscilloscope & Logic Analyzer

SystemVerilog, Artix-7, Vivado, Python

- Built a mixed-signal oscilloscope and 8-channel logic analyzer in SystemVerilog on an Artix-7 FPGA. Sampled analog input through the XADC and digital channels at the FPGA system clock, buffering captures in a circular pre-trigger BRAM.
- Designed analog and digital edge triggers and an FSM capture controller that preserves signal history from before the trigger. Added a streaming mode with on-chip decimation and min/max aggregation to reduce data over the UART link.
- Verified each RTL block with self-checking SystemVerilog testbenches in Vivado and closed timing at 100 MHz. Built a Python host application over a framed binary protocol for waveform display, measurements, and CSV export.

Flight Controller with Integrated ExpressLRS Receiver

KiCad, STM32, ExpressLRS, Betaflight

- Designed a 4-layer STM32 flight controller in KiCad with an IMU, blackbox flash, USB-C connectivity, and four DShot motor outputs. Created its Betaflight hardware configuration and resolved processor-resource conflicts.
- Integrated a 2.4 GHz ExpressLRS receiver and designed the antenna signal path with impedance matching, grounding, and RF layout controls for reliable communication.
- Designed a protected 2S-6S power system with a low-noise sensor supply, released fabrication files, assembled the board, and completed staged bring-up of power, communications, radio, and motor outputs.

SKILLS

PCB & EDA: Altium, schematic capture, PCB layout, multilayer, DRC/ERC, Gerber and BOM release, DFM.

Simulation & Test: LTspice, SPICE modeling and correlation, transient and AC analysis, MATLAB, test automation, failure analysis, root cause analysis, oscilloscope, logic analyzer, DMM, Digilent WaveForms SDK, JTAG/SWD, soldering and rework.

Embedded: STM32 (ARM Cortex-M), ESP32, MSP430, FreeRTOS, UART, I2C, SPI, CAN, USB, ADC/DAC, GPIO, interrupts, DMA.

Languages & Tools: C, C++, Python (NumPy, SciPy, Matplotlib, pySerial, Tkinter), Verilog, SystemVerilog, Vivado (FPGA/RTL), Assembly, Git, Grafana.